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Front-load deck for a stand-on mower

Abstract

In one embodiment, a stand-on mower, comprising a chassis having drive wheels operably coupled to the chassis, a platform arranged between the drive wheels, a front wheel frame, coupled to the chassis, and having pivot mounts and front wheels coupled to the pivot mounts, and a load deck mounted to the front wheel frame and having openings through which the pivot mounts extend.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
4878339	12/1988	Marier	180/209	A01D 34/64
7089722	12/2005	Laskowski	56/14.9	A01D 34/662
7448691	12/2007	Brooks	298/3	B60B 33/0049
9809263	12/2016	Mitchell	N/A	B62D 51/001
2009/0272088	12/2008	Losey	56/14.7	A01D 34/001
2021/0127577	12/2020	Weihl	N/A	A01D 34/66
2022/0379781	12/2021	Kucera	N/A	B62D 51/02

OTHER PUBLICATIONS

Website version before and proximal to Jun. 7, 2021—Swisher Versa mower:

<https://www.swisherinc.com/versa-389cc-30-rough-cut-tractor-vtrc30.html>. cited by applicant

Website version before and proximal to Jun. 7, 2021—Toro Model 74518 mower:

<https://www.toro.com/en/professional-contractor/commercial-mowers/grandstand-stand-on-mower-74518>. cited by applicant

Website version before and proximal to Jun. 7, 2021—Scag V-Ride II mower:

<https://www.scag.com/product/stand-on-mowers/v-ride-ii/>. cited by applicant

Website version before and proximal to Jun. 7, 2021—48" Gravely Pro-stance mower:

<https://www.gravely.com/en-us/power-equipment/stand-on-mowers/pro-stance>. cited by applicant

Website version before and proximal to Jun. 7, 2021—Encore stand-on mower:

<https://encoreequipment.com/mowers/rage/>. cited by applicant

Website version before and proximal to Jun. 7, 2021—Velke stand-on mower:

<https://web.archive.org/web/20210406191053/https://www.wrightmfg.com/>. cited by applicant

Website version before and proximal to Jun. 7, 2021—Ferris stand-on mower:

https://www.ferrismowers.com/na/en_us/products/stand-on-mowers.html. cited by applicant

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION (1) This application claims the benefit of U.S. Provisional Application No. 63/197,523, filed Jun. 7, 2021, which is hereby incorporated by reference in its entirety.

TECHNICAL FIELD

(1) The present disclosure is generally related to stand-on mowers.

BACKGROUND

(2) Lawn mowers are generally categorized as walk-behind, seated-riding, and stand-on mowers. Each type has their own perceived advantages and disadvantages, the choice of which depends on whether the use is through a commercial organization (e.g., lawn care business) or residential owner (e.g., private homeowner), cost constraints, and/or the frequency and/or type of application for which the mower is intended to be used. Stand-on mowers generally have a platform located between a pair of rear drive wheels that enables the operator to stand while operating the mower. Such mowers are often used by commercial organizations, though not limited as such, and may be chosen for any one or more of a variety of reasons. For instance, some perceived benefits of stand-on mowers (e.g., versus seated-riding) include ergonomics (e.g., reduced back strain, which is also a motivation for recent trends in standing desktops in office environments, and easier to mount), tighter operational and storage specifications (e.g., shorter length than seated riding mowers of the same cutting width permits improved maneuverability, and permits more machines to be loaded onto a trailer), and safety (e.g., easy to dismount in dangerous situations). With one or more of these perceived benefits, stand-on mowers have gained in popularity, yet design improvements are still needed to provide more versatility to these machines.

SUMMARY OF THE INVENTION

(3) In one embodiment, a stand-on mower, comprising a chassis having drive wheels operably coupled to the chassis, a platform arranged between the drive wheels, a front wheel frame, coupled to the chassis, and having pivot mounts and front wheels coupled to the pivot mounts, and a load deck mounted to the front wheel frame and having openings through which the pivot mounts extend.

(4) These and other aspects of the invention will be apparent from and elucidated with reference to the embodiment(s) described hereinafter.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Many aspects of a load deck for a stand-on mower of the present disclosure can be better understood with reference to the following drawings. The components in the drawings are not necessarily to scale, emphasis instead being placed upon clearly illustrating the principles of a load deck for a stand-on mower. Moreover, in the drawings, like reference numerals designate corresponding parts throughout the several views.

(2) FIGS. 1A-1B are schematic diagrams that illustrate, in front and rear isometric views, an embodiment of an example stand-on mower equipped with an embodiment of an example load deck.

(3) FIGS. 2A-2B are schematic diagrams that illustrate, in front and side isometric views, an embodiment of an example load deck used on an example stand-on mower.

(4) FIGS. 3A-3B are schematic diagrams that illustrate, in front isometric and elevation views, an example front wheel frame, upon which an embodiment of a load deck may be mounted.

(5) FIGS. 4A-4G are schematic diagrams that illustrate, in various views, example support members coupled to a lower surface of an embodiment of an example load deck and an example securement method among a front wheel frame, the support members, and the load deck.

(6) FIG. 5 is a schematic diagram that illustrates example dimensions and one example geometric shape of an embodiment of an example load deck.

DESCRIPTION OF EXAMPLE EMBODIMENTS

(7) Certain embodiments of a load deck and corresponding stand-on mower on which the load deck

is mounted are disclosed that provide for added utility or versatility to the normal mowing function of such machines. In one embodiment, the load deck is mounted to a front wheel frame and is a solid metal structure to carry loads that may be used for any lawn care or landscaping maintenance. For instance, the load deck may be used instead of a wheel barrel, landscaping dolly, or any type of push cart, and can be loaded with, for instance, a fifty-five (55) gallon or larger plastic container of mulch, landscape stone, sand, dirt, gravel, landscaping tools, among other material that is needed to facilitates lawn care/landscaping maintenance. The use of a load deck is beneficial for transporting a new installation or removal of existing landscaping to negate the need for other machines or equipment to carry such loads, such as trees, sod, shrubs, plants, paver bricks, large retaining wall blocks, garbage cans, branches, tree logs, rolls of landscaping straw, rolls of weedscreen, heavy landscape urns, and/or bags of fertilizer, which may lead to reduced cost and/or labor in the servicing of lawns and landscaping efficient operations.

(8) Digressing briefly, existing stand-on mowers lack a load deck, requiring the need for physically carrying bags, containers, and/or material or using an additional machine to carry such loads to the work site. In contrast, certain embodiments of a stand-on mower and load deck are disclosed, the installation of the load deck made possible through the efficient utilization of at least existing space on typical stand-on mowers by mounting (and securing) a load deck onto the front wheel frame, the load deck having sufficient structure to support the expected heavier loads often associated with landscape maintenance and related projects.

(9) Having summarized various features of certain embodiments of a stand-on mower with load deck of the present disclosure, reference will now be made in detail to the detailed description of a stand-on mower with load deck as illustrated in the drawings. While the disclosure is described in connection with these drawings, there is no intent to limit it to the embodiment or embodiments disclosed herein. For instance, though emphasis is placed on examples illustrating one type of model and manufacturer of a stand-on mower, it should be appreciated that the load deck may be arranged on other types of stand-on mowers, with perhaps some minor adjustments in dimensions and/or manner of securement for some model types to accommodate any variations in specifications, with these minor variations contemplated to be within the scope of the invention. Further, although the description identifies or describes specifics of one or more embodiments, such specifics are not necessarily part of every embodiment, nor are all various stated advantages associated with a single embodiment. On the contrary, the intent is to cover all alternatives, modifications and equivalents included within the scope of a stand-on mower with load deck as defined by the appended claims. Further, it should be appreciated in the context of the present disclosure that the claims are not necessarily limited to the particular embodiments set out in the description.

(10) FIGS. 1A-1B are schematic diagrams that illustrate, in front and rear isometric views, an embodiment of an example stand-on mower **10** equipped with an embodiment of an example load deck **12**. It should be appreciated by one having ordinary skill in the art that the example stand-on mower **10** illustrates one example stand-on mower using a particular manufacturer and model, and that in some embodiments, stand-on mowers of other manufacturers and/or model types may also be equipped in a similar arrangement with the example load deck **12** and hence are contemplated to be within the scope of the invention. The stand-on mower **10** comprises conventional components, including a chassis or frame **14** comprising plural (e.g., a pair) of rear drive wheels **16** operably coupled to the chassis **14**, and a platform **18** arranged between the rear drive wheels **16**. The platform **18** is arranged centrally and at the rear of the stand-on mower **10**, enabling an operator to control the movement of the stand-on mower **10** through manipulation of the mower controls **20**. Also mounted to the chassis **14** is a fuel reservoir **22**, engine **24**, and a mower (or cutter) deck **26**. Engine and motor deck operation are controlled by the operator using the mower controls **20**, as is known. For instance, an operator may raise and lower the mower deck **26** using the mower controls **20**. The stand-on mower **10** further comprises the load deck **12** at the front of the stand-on mower

10, which is mounted to pivot mount assemblies to which the front wheels **13** are coupled. Note that the load deck **12** is used in addition to the mower deck **26**, and is not a decorative panel or covering from other components of the stand-on mower **10**, but rather, a load-bearing structure (e.g., carrying loads of approximately five hundred (500) pounds or more). As explained below in association with FIGS. **3A-3B**, the load deck **12** is mounted to a front wheel frame, and is used to transport tools, equipment, and/or material to a work location. For instance, an operator may place tools, bags of stone or containers of fertilizer or weed killer, etc., on the load deck **12** for transport to a desired work-site location.

(11) Referring to FIGS. **2A-2B**, shown are front and side isometric views of one embodiment of the load deck **12**. In some embodiments, the load deck **12** may take on a different geometric configuration and/or structure. Note that discussion of like-numbered structures for the same structures shown in FIGS. **1A-1B** (and other figures described herein) is omitted here for brevity, with emphasis hereinafter on the load deck **12**. The load deck **12** is shown with a first portion **28** comprising an upper surface **30** (e.g., upper-facing) and lower surface (ground-facing, as shown in FIG. **4A**). In one embodiment, the load deck **12** comprises a second portion **32** that joins (e.g., is coupled to, including integrated) with the first portion **28**. In one embodiment, the second portion **32** is upright (e.g., orthogonal to the first portion **28**). In some embodiments, the second portion **32** is substantially orthogonal (e.g., within approximately ± 0.1 -10 degrees from the upright position). In one embodiment, the second portion **32** is formed by bending one end (opposite the apex end) of the load deck upward. In some embodiments, a separate piece of material (e.g., of metal or other material) used as the second portion **32** may be welded or more generally, secured, to the first portion **28**. In some embodiments, the first portion **28** and second portion **32** may be cast or forged as a unitary piece in the L-shaped configuration as shown in FIGS. **2A-2B**. In some instances, the second portion **32** may function to prevent tipping during transport. In some embodiments, the second portion **32** may be omitted. In one embodiment, the upper surface of the first portion **28** may comprise a stippled surface (e.g., rough surface). For instance, the stipples (e.g., protrusions) may provide a friction surface to reduce the risk of loads on the load deck **12** from sliding off or sliding off-center of the load deck **12** (e.g., to prevent causing an imbalance). The stipples may be arranged according to a regular pattern, as shown in FIGS. **2A-2B**, or arranged as an irregular pattern. In some embodiments, the stipples may be located in only a sub-portion of the upper surface **30** of the load deck **12**, or in some embodiments, may be omitted. In some embodiments, dimples may be used additionally or as an alternative to stipples.

(12) The second portion **32** comprises a forward-facing surface **34** and a rearward-facing surface **36**. The use or omission of stipples in all or a portion of the forward-facing surface **34** is as similarly described for the upper surface **30** of the first portion **28**, and hence omitted here for brevity. The lower surface of the first portion **28** and the rearward-facing surface **36** may be comprised of a smooth surface, though in some embodiments, stipples (and/or dimples) may be arranged or omitted as described above for upper surface **30** and forward-facing surface **34** may likewise be used similarly for the lower surface of the first portion **28** or the rearward-facing surface **36**.

(13) In one embodiment, the load deck **12** is comprised of a metal material (e.g., steel). In one embodiment, the load deck **12** is approximately 1/16^{sup.th} inch thickness, but other thicknesses may be used depending on the intended loads to be carried. In some embodiments, other material may be used for the load deck **12**, or portions of the load deck **12**. For instance, the second portion **32** may be comprised of a non-metal material with sufficiently high strength and durability for carrying loads.

(14) Before explaining further the structure and manner of mounting the load deck according to various embodiments, attention is directed to FIGS. **3A-3B** to show an example front wheel frame **38** coupled to the chassis **14** and to which the load deck **12** is mounted. For instance, FIGS. **3A-3B** reveal the front wheel frame **38** before the load deck **12** is mounted to the front wheel frame **38**. As

shown, the front wheel frame **38** comprises a pair of pivot mount assemblies **40** with forwardly extending frame members **42** and a laterally-oriented frame member **44** arranged between and secured (e.g., welded) to the forwardly extending frame members **42**. One end of each of the forwardly extending frame members **42** is coupled to the chassis **14**. For instance, the coupling shown in FIGS. 3A and 3B are bolted flanged assemblies of the chassis **14** and the forwardly extending frame members **42**, though other mechanisms of securement may be used in some embodiments. At the other end of each of the forwardly extending frame members **42** are respective pivot mounts **46** that are each coupled to the front wheels **13**. Note that the forwardly extending frame members **42** are shown as slightly angled fore-and-aft relative to a longitudinal centerline of the stand-on mower **10**, though in some embodiments, the members **42** may be parallel to the centerline or at a different angle relative to the longitudinal centerline than shown. The front wheels **13** comprise caster wheels, as is typical of stand-on mowers, though other types of front wheel and steering mechanisms may be used in some embodiments. Also shown are mower deck wheels **48** of the mower deck **26**, though discussion of the same is omitted here for clarity and to avoid obfuscating relevant features of the invention. As is discussed further below, the load deck **12** is mounted to the front wheel frame **38**, with securement (e.g., welding) of the load deck **12** to the laterally-oriented frame member **44**, the forwardly extending frame members **42**, and the pivot mounts **46**.

(15) Attention is now directed to FIGS. 4A-4G, which illustrate example support members **50** coupled to a lower surface **52** of the load deck **12**, and which further illustrate an example method of securement (e.g., welds) among the front wheel frame **38**, the support members **50**, and the load deck **12**. Note that in some embodiments, fewer or additional support member **50** may be used. Though the support members **50** are shown arranged in a triangular-shaped arrangement (with the apex at or adjacent to the apex of the rounded end of the load deck **12**), in some embodiments, the manner of arrangement of the support members **50** may be different than the triangular configuration shown. In some embodiments, the support members **50** may be omitted, depending on the intended application, budgetary constraints, carrying capacity, and/or thickness of the load deck **12**, among other factors. For instance, the load deck **12** may be thicker (e.g., greater than 1/16 sup.th inch) and/or the anticipated loads may be of less weight than, say, approximately five-hundred (500) pound rating intended for the present design. The support members **50** help prevent or mitigate deformation of the load deck **12** under load and/or loads over time (e.g., the expected life of the stand-on mower **10**). Note that some reference lines and numbers for other features of the stand-on mower **10**, previously described and depicted, are omitted here to avoid obfuscating relevant features. With particular focus on FIGS. 4A-4D, the support members **50** are depicted as L-shaped, metal (e.g., iron) support beams coupled to the lower surface **52** of the load deck **12** and arranged at a point or apex centrally adjacent the rounded, forward-most end of the load deck **12** and extended rearwardly from the apex to each of the forwardly extending frame members **42** at locations that are adjacent to, and rearward of, each of the pivot mounts **46**. In some embodiments, the support members **50** may be embodied according to a different form (e.g., square tubes, rounded supports, etc.). In one embodiment, the support members **50** are welded to the lower surface **52** of the load deck **12** at plural, spaced-apart locations as represented in a few locations by welds **54A**. The (e.g., upright) ends of the support members **50** adjacent the respective pivot mounts **46** are welded to the forwardly extending frame members **42**, as shown illustratively by welds **54B**. Note that additional or fewer locations for the welds **54** may be used in some embodiments, and/or in some embodiments, the coverage area of each of the welds **54** may be less or greater. In some embodiments, the location where the ends of the support members **50** are secured (e.g., welded) to the forwardly extending frame members **42** may be different. Note that other and/or additional mechanisms/methods of securement may be used, including for instance, through the use of bolts, screws, etc.

(16) Referring to FIG. 4E, shown is one of the pivot mounts **46**, representative of each of the pivot

mounts **46**, and further illustrates how the load deck **12** comprises an opening **56** (e.g., notch) that permits the pivot mount **46** to extend above and below a plane of the load deck **12**. That is, the pivot mount **46** extends above and below the load deck **12**, and in one embodiment, the load deck **12** is secured to the pivot mount **46** via a weld **54C** (and in some embodiments, also welded between the load deck **12** and this pivot mount **46** on the other side that is not shown in this view). Similar weld(s) are implemented for the opposing side pivot mount **46**. In some embodiments, the opening **56** may fully surround the pivot mount **46** (versus partially surrounding), such as if the pivot mounts **46** were not angled or angled as much from the longitudinal centerline of the stand-on mower **10**, or if the load deck **12** was widened, or for other design reasons.

(17) FIG. **4F** is a close-up view of a lower portion of the rearward-facing surface **36** of the second (e.g., upright) portion **32** of the load deck **12**. In particular, an example weld **54D** is shown that secures the second portion **32** to the laterally-oriented frame member **44** of the chassis **14**. It should be appreciated that additional welds may be used along the junction between the lower portion of the second portion **32** and the laterally-oriented frame member **44**. In some embodiments, there may be plural welds **54D** spaced apart, or in some embodiments, the weld coverage may be extended (e.g., along a greater lateral distance and/or entirely along the junction).

(18) FIG. **4G** is a side-perspective view that shows an example weld **54E** that secures the lower surface **52** of the first portion **28** of the load deck **12** to one of the forwardly extending frame members **42** at a location proximal to the end of the load deck **12** that joins with the (e.g., upright) second portion **32**. As explained similarly above, though one weld **54E** is shown, it should be appreciated that additional welds **54E** and/or different weld coverage areas per weld may be used in some embodiments. It should be appreciated that a similar securement is achieved between the load deck **12** and the other forwardly extending frame members **42**.

(19) Referring now to FIG. **5**, shown is schematic diagram that illustrates example dimensions and geometric shape of the load deck **12**. It should be appreciated that the dimensions depicted in FIG. **5** are merely illustrative of a load deck **12** for a particular manufacturer and model type of stand-on mower **10**, and that in some embodiments, different dimensions/specifications may be used for the same manufacturer/model or different manufacturers/model types. The diagram depicts a top plan view, with the support members **50** (shown in phantom, dashed line) forming (at the lower surface of the load deck **12**) a triangular shape that meets at an apex **58** coincident with a longitudinal line or plane **60** that runs through the apex **62** of the forward most location of the rounded end of the load deck **12**. In other words, the apexes **58** and **62** are proximal to one another, and on the same longitudinal plane **60**. As shown, the load deck **12** is shaped in a substantially semi-circular form, though other geometries may be used in some embodiments. The openings **56** (e.g., notch openings) permit the pivot mounts **46** (not shown in FIG. **5**) to be positioned to extend above and below the load deck **12**. Further, the load deck **12** clearly extends substantially beyond the pivot mounts **46**, offering a generous surface area for carrying various types and dimensions of loads. In one embodiment, the dimension “A” corresponds to a distance from the rearward facing surface **36** of the second portion **32** to the rear-most location of the opening **56**, and the dimension “B” corresponds to a distance from the rearward facing surface **36** of the second portion **32** to the forward-most location of the opening **56** of the second portion. In one embodiment, the dimensions are at or approximately 7 inches for “A”, and at or approximately 10 inches for “B”. The dimension “C” corresponds to a diameter of the semi-circular shape of the first portion **28**, and in one embodiment, is at or approximately 69 inches. As to the second portion **32**, in one embodiment, dimensions are at or approximately 8 inches in height, 36 inches in width (side-to-side), and ½ inch thickness. As explained above, these dimensions are merely for illustration, and other dimensions are contemplated to be within the scope of the disclosure.

(20) In view of the above description, it should be appreciated by one having ordinary skill in the art that one embodiment of a method of installing a load deck for a stand-on mower comprises forming a load deck for a stand-on mower, the stand-on comprising a chassis having drive wheels

operably coupled to the chassis, a platform arranged between the drive wheels, a front wheel frame, coupled to the chassis, and having pivot mounts and front wheels coupled to the pivot mounts, and a mower deck arranged between the front wheels and the drive wheels. For instance, forming the load deck may comprise cutting the steel from a larger flat, steel sheet into a substantially semi-circular, flat metal sheet of suitable thickness, along with openings for enabling a suitable fit with the pivot mounts. Note that the semi-circular shape is one geometric design, and that in some embodiments, the metal sheet may be formed of other geometric configurations (e.g., rectangular or other multi-sided configurations). In some embodiments, the cutting may be achieved using computer-controlled machinery or performed manually or a combination of both. The cutting may be performed using a blade (e.g., diamond blade or other metal cutting or abrasive blades for cutting steel), laser, milling machine, or other metal cutting techniques known in the art. In some embodiments, the metal plate may be cast or forged. In some embodiments, the metal plate may be patterned (e.g., with stipples or otherwise to roughen the surface(s)) on all or a portion of one side or both sides using known techniques, or in some embodiments, the raw metal material may be received as a patterned sheet. In some embodiments, patterning may be omitted.

(21) In some embodiments where a back plate (e.g., second portion or upright plate) is used, the method further comprises forming the second portion. In one embodiment, the second portion is formed by bending the formed metal plate, resulting in the flat metal first portion and the substantially upright second portion. In some embodiments, the second portion is formed similarly to the formation of the metal plate (e.g., a first portion) and affixed (e.g., via welding, bolts, etc.) the second portion to the first portion.

(22) In some embodiments, the first and second portions are formed at once (e.g., via a forging or cast operation).

(23) In some embodiments, patterning may further be applied or formed during the upright plate forming operation, or omitted in some embodiments.

(24) In some embodiments, the method further comprises affixing one or more support members to the lower surface of the load deck. In one embodiment, the support members are arranged in a triangular arrangement and welded at spaced apart locations or continuously at support member surfaces adjacent the lower surface of the metal plate (e.g., first portion). In some embodiments, a single support member may be welded to the lower surface, or in some embodiments, two or more support members may be welded to the lower surface according to any of a plurality of different geometric arrangements suitable for providing support for loads of the desired capacity placed on the load deck. In some embodiments, the support members may be attached using other affixing mechanisms (e.g., screws, bolts, etc.). In some embodiments, depending on the strength and/or thickness of the metal plate, the support members may be omitted.

(25) The method further comprises mounting the load deck to the front wheel frame and forward of the mower deck. For instance, the load deck may be welded in spaced apart locations, or continuously, between the lower surface of the load deck and the front wheel frame, as explained above. In some embodiments, the load deck may be affixed at least in part to the front wheel frame through welds between the support members and the front wheel frame. In some embodiments, there may be additional welds between the upper surface of the load deck and the pivot mounts. Note that welds may be achieved using automated or semi-automated welding machines, or in some embodiments, performed manually or a combination of both techniques. In some embodiments, other and/or additional affixing mechanisms may be used (e.g., screws, bolts, etc.) for mounting the load deck onto the front wheel frame.

(26) Any process descriptions described for the aforementioned method should be understood as representing steps in a process, and alternate implementations are included within the scope of the embodiments in which steps may be executed out of order from that discussed, including substantially concurrently, depending on the functionality involved, as would be understood by those reasonably skilled in the art of the present disclosure.

(27) While the invention has been illustrated and described in detail in the drawings and foregoing description, such illustration and description are to be considered illustrative or exemplary and not restrictive; the invention is not limited to the disclosed embodiments. Other variations to the disclosed embodiments can be understood and effected by those skilled in the art in practicing the claimed invention, from a study of the drawings, the disclosure, and the appended claims. Note that various combinations of the disclosed embodiments may be used, and hence reference to an embodiment or one embodiment is not meant to exclude features from that embodiment from use with features from other embodiments. In the claims, the word “comprising” does not exclude other elements or steps, and the indefinite article “a” or “an” does not exclude a plurality.

Claims

1. A stand-on mower, comprising a chassis having drive wheels operably coupled to the chassis, a platform arranged between the drive wheels, a front wheel frame, coupled to the chassis, and having pivot mounts and front wheels coupled to the pivot mounts, a mower deck arranged between the front wheels and the drive wheels, and a load deck mounted to the front wheel frame and having openings through which the pivot mounts extend, the load deck arranged fore and aft beginning from proximal to a front end of the mower deck to beyond the pivot mounts and having an upper-facing, load bearing surface with sufficient area to enable transport of lawncare, or landscape maintenance, material.
 2. The stand-on mower of claim 1, wherein the load deck comprises a first portion having the upper-facing, load bearing surface and a lower surface, wherein the pivot mounts extend above and below the first portion.
 3. The stand-on mower of claim 2, further comprising one or more support members coupled to the lower surface of the load deck and to the front wheel frame, and wherein the one or more support members are secured to the lower surface of the load deck and to the front wheel frame.
 4. The stand-on mower of claim 2, wherein the load deck is secured to the pivot mounts.
 5. The stand-on mower of claim 2, wherein the load deck is secured to the front wheel frame.
 6. The stand-on mower of claim 2, wherein the load deck further comprises a second portion that joins the first portion and that is substantially orthogonal to the first portion.
 7. The stand-on mower of claim 2, wherein the first portion comprises a substantially semi-circular shape.
 8. The stand-on mower of claim 1, wherein the load deck comprises a steel material having a thickness of approximately 1/16 sup.th inch or more.
 9. The stand-on mower of claim 1, wherein the upper-facing, load bearing surface comprises a rough surface.
 10. The stand-on mower of claim 9, wherein the rough surface comprises a stippled surface.
 11. The stand-on mower of claim 1, wherein the front wheels comprise caster wheels.
 12. A method of installing a load deck for a stand-on mower, comprising: forming a load deck for a stand-on mower, the stand-on mower comprising a chassis having drive wheels operably coupled to the chassis, a platform arranged between the drive wheels, a front wheel frame, coupled to the chassis, and having pivot mounts and front wheels coupled to the pivot mounts, and a mower deck arranged between the front wheels and the drive wheels; and mounting the load deck to the front wheel frame, the load deck arranged fore and aft beginning from proximal to a front end of the mower deck to beyond the pivot mounts and having an upper-facing, load bearing surface with sufficient area to enable transport of lawncare, or landscape maintenance, material.
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